

ArchiMate® 3.2 – Modeling Standard for Enterprise Architecture



Business Layer

These elements describe products and services which a company provides externally. The views show how these products and services are realized.

Business Actor

Represents a business entity that is capable of performing behavior.

Business Event

Represents a business-related state change.

Business Function

Represents a collection of business behavior based on a chosen set of criteria such as required business resources and/or competencies, and is managed or performed as a whole.

Business Interaction

Represents a unit of collective business behavior performed by (a collaboration of) two or more business actors, business roles, or business collaborations.

Business Collaboration

Represents an aggregate of two or more business internal active structure elements that work together to perform collective behavior.

Business Object

Represents a concept used within a particular business domain.

Business Process

Represents a sequence of business behaviors that achieves a specific result such as a defined set of products or business services.

Business Role

Represents the responsibility for performing specific behavior, to which an actor can be assigned, or the part an actor plays in a particular action or event.

Business Interface

Represents a point of access where business services are made available to the environment.

Business Service

Represents explicitly defined behavior that a business role, business actor, or business collaboration exposes to its environment.

Product

Represents a coherent collection of services and/or passive structure elements, accompanied by a contract, which is offered as a whole to (internal or external) customers.

Representation

Represents a perceptible form of the information carried by a business object.

Contract

Represents a formal or an informal specification of an agreement between a provider and a consumer that specifies the rights and obligations associated with a product and establishes functional and non-functional parameters for interaction.

Application Layer

Applications which support the business layer are portrayed in the application layer.

Application Event

Represents an application state change.

Application Function

Represents automated behavior that can be performed by an application component.

Application Interaction

Represents a unit of collective application behavior performed by (a collaboration of) two or more application components.

Application Collaboration

Represents an aggregate of two or more application internal active structure elements that work together to perform collective application behavior.

Application Component

Represents an encapsulation of application functionality aligned to implementation structure, which is modular and replaceable.

Application Process

Represents a sequence of application behaviors that achieves a specific result.

Application Interface

Represents a point of access where application services are made available to a user, another application component, or a node.

Application Service

Represents an explicitly defined exposed application behavior.

Data Object

Represents data structured for automated processing.

Technology Layer

Infrastructure which consists of hardware and software needed to implement applications are described in the technology layer.

Artifact

Represents a piece of data that is used or produced in a software development process, or by deployment and operation of an IT system.

Device

Represents a physical IT resource upon which system software and artifacts may be stored or deployed for execution.

Node

Represents a computational or physical resource that hosts, manipulates, or interacts with other computational or physical resources.

Communication Network

Represents a set of structures that connects devices or system software for transmission, routing, and reception of data.

Path

Represents a link between two or more technology internal active structure elements, through which these elements can exchange data, energy, or material.

System Software

Represents software that provides or contributes to an environment for storing, executing, and using software or data deployed within it.

* Generic elements have not been included to keep things clearer. Generic elements also appear in the technology layer.

Physical Layer

The physical layer adds further physical components to the technology layer, such as manufacturing facilities and buildings.

Distribution Network

Represents a physical network used to transport materials or energy.

Facility

Represents a physical structure or environment.

Equipment

Represents one or more physical machines, tools, or instruments that can create, use, store, move, or transform materials.

Material

Represents tangible physical matter or energy.

General Elements

General elements can be used across all layers.

Grouping

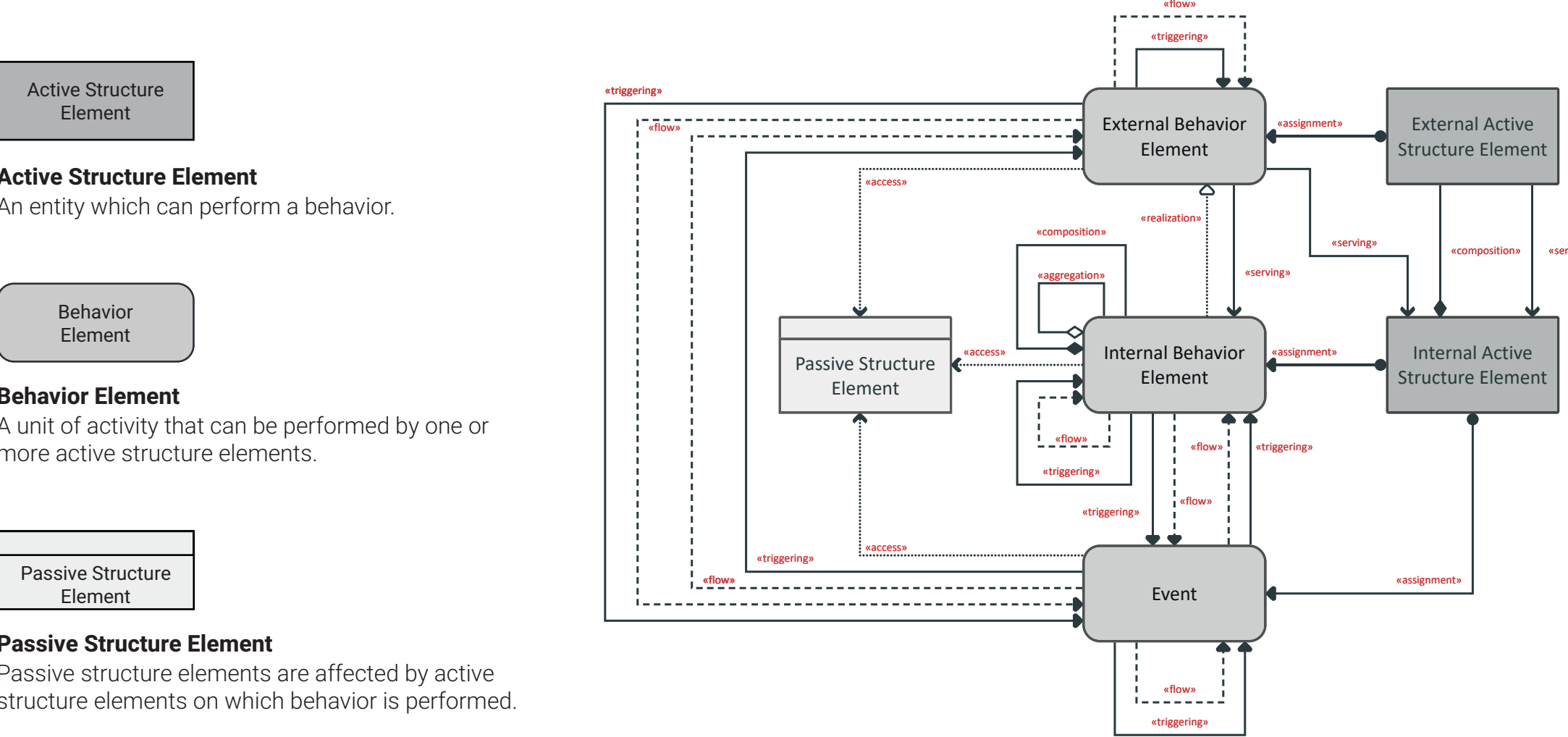
Aggregates or composes concepts that belong together based on some common characteristic.

Location

Represents a conceptual or physical place or position where concepts are located (e.g., structure elements) or performed (e.g., behavior elements).

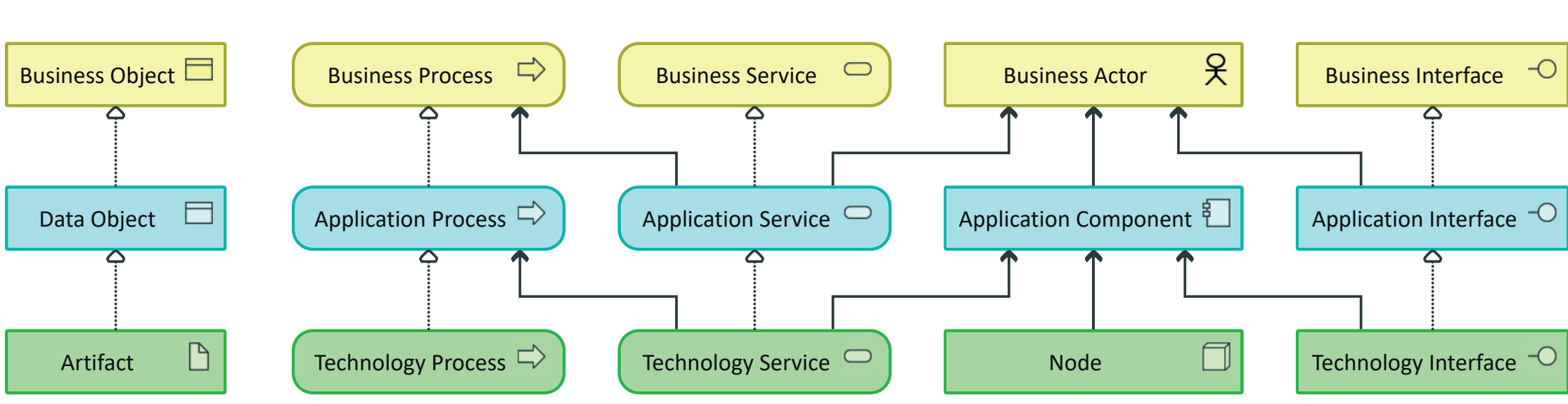
Generic Metamodel

The generic metamodel is a repetitive construct that can be used in every architecture layer. It describes the possible relationships within a layer.



Business-IT Alignment

The view depicted shows various relationships between the business layer, application layer and technology layer elements. There are generally two types of relationships between the layers. This representation does not show all relationships and elements possible in the layers in ArchiMate®.



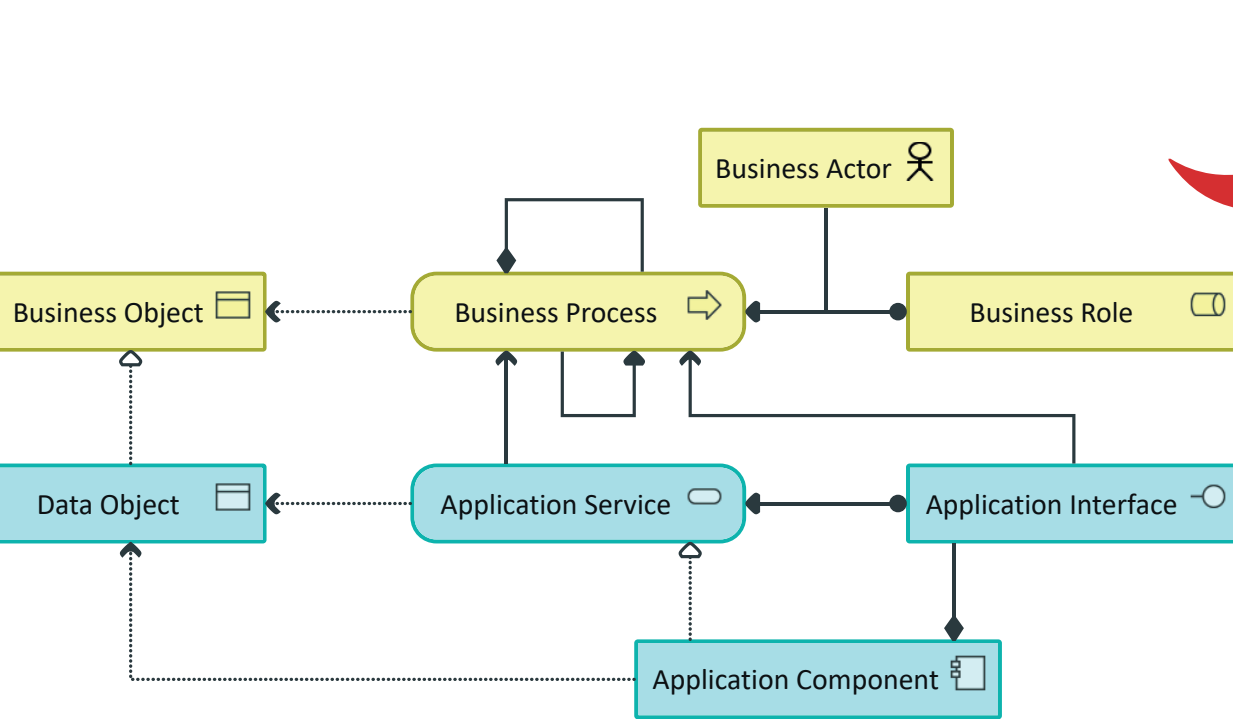
Viewpoint Description

The following aspects are described for each viewpoint.

Stakeholder	Enterprise, process and application architectures, operation manager
Concerns	Consistency and completeness, reduction of complexity
Purpose	Design, decision making
Abstraction Level	Relationships
Layers	Business and Application layers
Aspects	Behaviors and structures

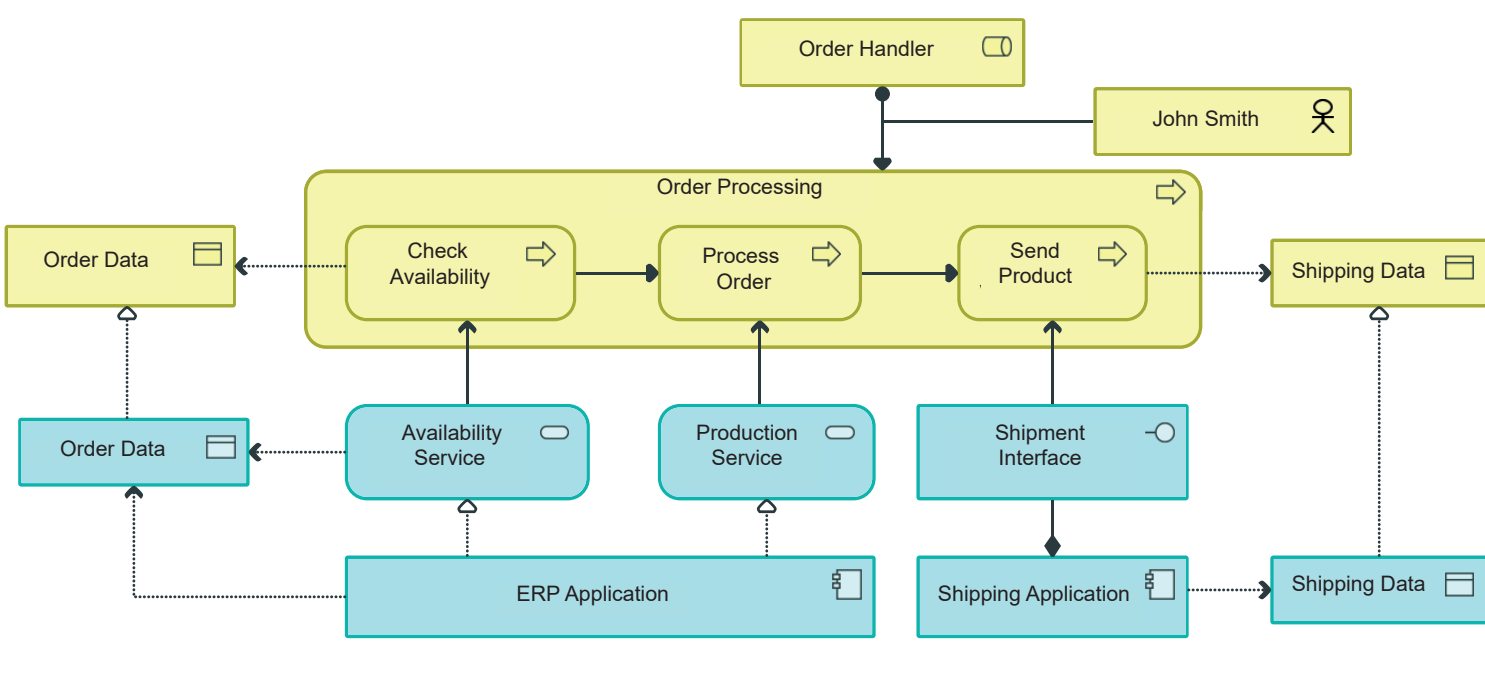
Viewpoint

Subset of the metamodel that defines a view of the Enterprise Architecture for a certain purpose. Example for a predefined Application Usage Viewpoint in ArchiMate®.



View

View from the Enterprise Architecture to a certain point in time defined by a viewpoint. Example of an Application Usage View for process architects in the "Order Processing" process and application architects to visualize process automation.



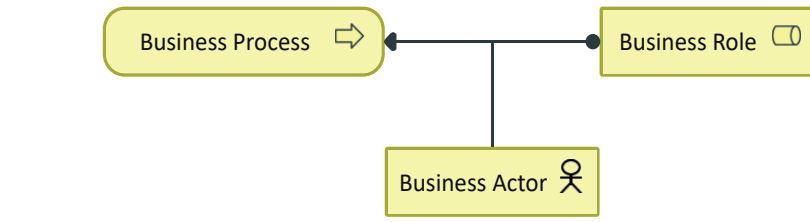
ArchiMate® Layers

	Passive Structure	Behavior	Active Structure	Motivation
Strategy				
Business				
Application				
Technology				
Physical				
Implementation & Migration				

Aspects

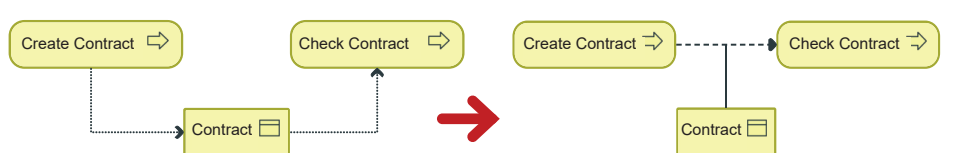
Roles & Responsibilities

A ternary relationship (association based on assignment) can be used to model roles and responsibilities.



Alternative Representations

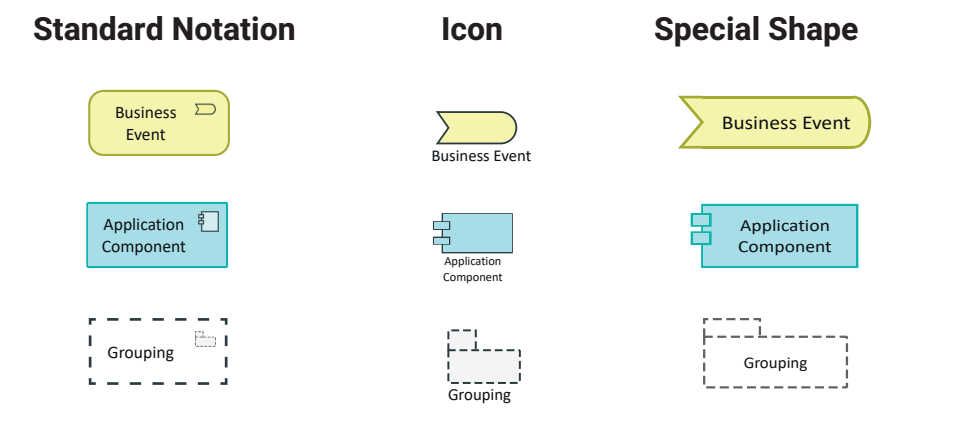
Data transfer can either be represented as two accesses or one association in a flow.



Aggregation, composition and assignment can also be represented with nesting of one element in another.



Elements can be displayed in three different ways: Standard notation, only as an icon or as a special shape.



Motivation Elements

The Motivation Extension enables decisions to changes in the architecture to be justified and to make influences considerably clearer.

Requirement

Represents a statement of need defining a property that applies to a specific system as described by the architecture.

Meaning

Represents the knowledge or expertise present in, or the interpretation given to, a concept in a particular context.

Stakeholder

Represents the role of an individual, team, or organization (or classes thereof) that represents their interests in the effects of the architecture.

Assessment

Represents the result of an analysis of the state of affairs of the enterprise with respect to some driver.

Principle

Represents a statement of intent defining a general property that applies to any system in a certain context in the architecture.

Constraint

Represents a limitation on aspects of the architecture, its implementation process, or its realization.

Outcome

Represents an end result, effect, or a consequence of a certain state of affairs.

Driver

Represents an external or internal condition that motivates an organization to define its goals and implement the changes necessary to achieve them.

Value

Represents the relative worth, utility, or importance of a concept.

Goal

Represents a high-level statement of intent, direction, or desired end state for an organization and its stakeholders.

Strategy Elements

The architecture can be viewed from a strategic viewpoint with strategic elements.

Capability

Represents an ability that an active structure element, such as an organization, person, or system, possesses.

Course of Action

Represents an approach or plan for configuring some capabilities and resources of the enterprise, undertaken to achieve a goal.

Resource

Represents an asset owned or controlled by an individual or organization.

Value Stream

Represents a sequence of activities that create an overall result for a customer, stakeholder, or an end user.

Implementation & Migration Elements

Implementation & migration elements are a graphical description of the architecture project or development and help with the implementation.

Gap

Represents a statement of difference between two plateaus.

Deliverable

Represents a precisely-defined result of a work package.

Work Package

Represents a series of actions identified and designed to achieve specific results within specified time and resource constraints.

Implementation Event

Represents a state change related to implementation or migration.

Plateau

Represents a relatively stable state of the architecture that exists during a limited period of time.

Risk & Security Overlay

An overlay has been developed for integration of risk and security aspects in the architecture model.

Threat Event

Event with the potential to adversely impact an asset. An attack is a specific type of threat event that is the result of an intentional malicious activity of an attacker, which is a specific type of threat agent.

Control Measure

An action, device, procedure, or technique that reduces a threat, a vulnerability, or an attack by eliminating or preventing it, by minimizing the harm it can cause, or by discovering and reporting it so that corrective action can be taken.

Risk

The probable frequency and probable magnitude of future loss.

Risk Domain

A domain consisting of entities that share one or more characteristics relevant to risk management or security. A risk domain is also a context or set of conditions that affects a risk exposure level.

Loss Event

Any circumstance that causes a loss or damage to an asset.

Control Objective

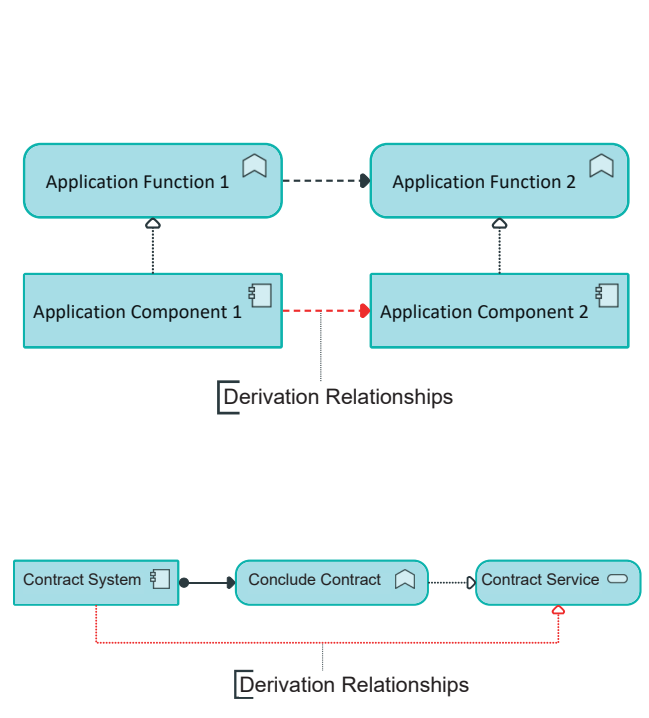
Aim or purpose of specified control measures which address the risks that these control measures are intended to mitigate.

Vulnerability

The probability that an asset will not be able to resist the actions of a threat factor.

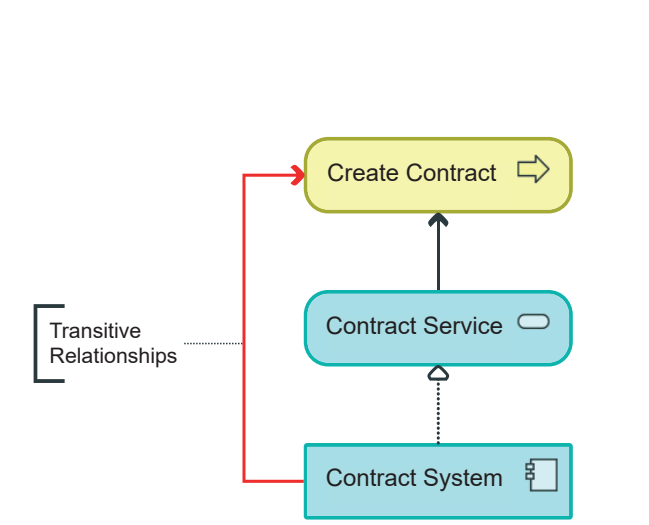
Derivation Relationships

Dynamic relationships can be transferred to assigned structure elements or realized services.



Transitive Relationships

If elements have a structural relationship with each other or are in structural relationships and dependency relationships, then they can be combined and the weakest relationship can be used as the derived relationship.



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